Hierarchical Prototype Learning via Aggregation-Decomposition for Fine-Grained Geospatial Scene Graph Generation

Linlin Wang, Tingzhu Wang, Junwei Luo, and Yansheng Li, *Senior Member, IEEE*

***Abstract*—Geospatial scene graph generation (G-SGG) in remote sensing imagery (RSI) aims to identify triplets <subject, relationship, and object> that reflect interactions among objects on the Earth's surface. A key challenge in G-SGG is the large intraclass variation of geospatial relationships, which leads to frequent misclassification. To address this, we propose a hierarchical prototype learning network (HPL-Net) via aggregation-decomposition, designed to enhance fine-grained G-SGG by addressing intraclass variation in geospatial relationships. In HPL-Net, hierarchical constraint (HC) of prototype learning, employing a tree structure, is designed to learn discriminative representations of relationship prototypes. In addition, a class-balanced constraint (CC) of instance learning with class-aware margins is introduced to mitigate misclassification caused by imbalanced relationship categories. Extensive experiments demonstrate that our method outperforms state-of-the-art G-SGG models on the satellite-based STAR and drone-based AUG datasets. Specifically, it outperforms the latest models, such as RPCM and LPG, by significant margins of 4.90%/5.32% and 4.52%/4.31% on TMR@1500/2000 and F@50/150, respectively, for the predicate classification (PredCls) task on the STAR and AUG datasets. The code will be available at https://github.com/linlin-dev/HPL-Net**

***Index Terms*—Aggregation-decomposition, fine-grained geospatial relationship, geospatial scene graph generation (G-SGG), hierarchical prototype learning, remote sensing imagery (RSI).**

I. Introduction

OBJECTS on Earth's surface do not exist as isolated entities; instead, they are interconnected and form an integrated whole. Unlike scene graph generation (SGG) in natural imagery, geospatial SGG (G-SGG) in remote sensing imagery (RSI) aims to detect objects (spatially intersecting or disjoint) on Earth's surface and predict the geospatial relationships between objects, which has significant potential for cognitive interpretation tasks, such as image description [1] and visual question answering [2], as well as practical applications including traffic planning [3] and situation awareness [4].

Fine-grained geospatial relationship understanding is essential for capturing nuanced object interactions and comprehensively interpreting complex geospatial scenarios. RSI from satellites or drones, with its extensive spatial coverage and high-resolution details, provides crucial support for fine-grained G-SGG. Although traditional perceptual understanding tasks [5], [6], [7] in RSI have made remarkable progress in recognizing individual objects or regions, they have limited capacity to model complex interactions among multiple entities. In contrast, G-SGG captures structured interactions between object pairs, enabling a deeper interpretation of scene semantics. A key challenge in fine-grained geospatial relationship understanding, an essential component of G-SGG in RSI, is the significant intraclass variation among subject-object pairs, which often leads to relationship misclassification. For instance, as shown in Fig. 1(c), the samples <ship, approach, and dock> and <truck, approach, and toll_gate> have the same relationship category "*approach*," but exhibit substantial visual dissimilarity. Prototype-based approaches [8], [9], [10] have been applied in natural imagery to improve classification accuracy by learning compact and discriminative relationship representations in the latent space. In RSI, prototype learning has also been explored for tasks, such as classification [11], [12], segmentation [13], [14], and object detection [15], [16], demonstrating its potential to learn discriminative representations in complex geospatial scenarios. However, for the G-SGG task, the same relationship category can manifest diverse representations due to variations in subjects and objects, making it challenging for these methods to effectively capture such variability. As a result, their ability to accurately predict fine-grained geospatial relationships is significantly constrained.

An intriguing phenomenon in the G-SGG dataset from RSI is the natural hierarchical structure of relationship prototypes. As shown in Fig. 1(b) and (c), each relationship prototype, exhibiting intraclass variation, can be decomposed downward into multiple **child prototypes**. For instance, the relationship prototype "*violently emit (ve)*" can be decomposed into "ve_1" to "ve_m." Furthermore, similar relationship prototypes can be aggregated upward into **parent prototypes**. For example, the samples corresponding to ve and "*slightly emit (se)*" exhibit

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The authors are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: wangll@whu.edu.cn; tingzhu.wang@whu.edu.cn; luojunwei@whu.edu.cn; yansheng.li@whu.edu.cn).

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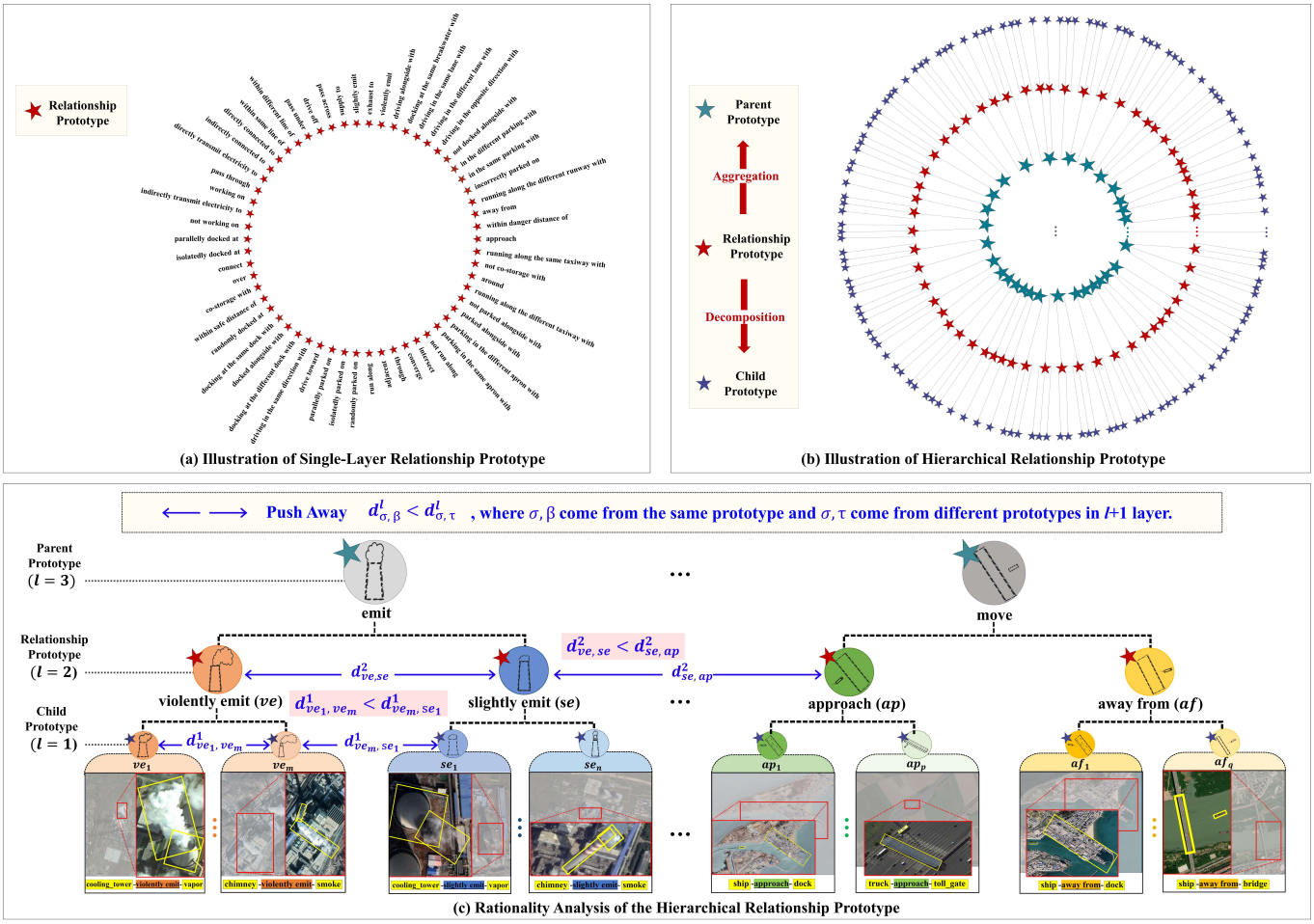


Fig. 1. Rationality analysis of our hierarchical relationship prototype compared with the traditional single-layer relationship prototype. (a) Single-layer relationship prototype indiscriminately pushes other prototypes away, overlooking hierarchical structures and intraclass variations, thereby compromising its effectiveness. (b) Our approach introduces down-decomposition to capture fine-grained variations of geospatial relationships and up-aggregation to enforce HCs. This alignment with real-world patterns substantially improves the interpretability and accuracy of fine-grained geospatial relationship classification. (c) Child prototype count varies with intraclass variation, and HCs ensure that prototypes originating from the same parent prototype are positioned closer together than those from different parents.

visual similarity and share the same parent prototype "*emit*." By decomposing into multiple child prototypes, the hierarchy models intraclass variation in geospatial relationships, capturing fine-grained distinctions in complex scenarios. Each child prototype is constrained by its corresponding parent prototype, regulating the distribution of prototypes in the latent space. This hierarchical structure aligns with real-world patterns, enhancing both interpretability and accuracy in geospatial relationship classification. However, traditional SGG methods [8], [17] often model relationship categories as single-layer relationship prototypes [Fig. 1(a)], indiscriminately pushing all other prototypes as negatives, while ignoring hierarchical structures and intraclass variation.

To tackle the above challenges, this article proposes a hierarchical prototype learning network (HPL-Net) via aggregation-decomposition for G-SGG in RSI. In the HPL-Net, hierarchical relationship prototypes and discriminative relationship representations are dynamically optimized and matched through hierarchical constraint (HC) of prototype learning and class-balanced constraint (CC) of instance learning. Guided by this hierarchical structure, HC is designed to enhance the distinguishability between prototypes and mitigate overlap between prototype spaces. CC is used to select high-quality positive and negative samples, ensuring that intraclass representations are more compact while pushing away negative samples. To avoid tail-class misclassification caused by class imbalance, CC introduces a class-aware margin factor to expand the prototype space of tail classes, achieving accurate classification.

The contributions can be summarized as follows.

1) We propose a HPL-Net via aggregation-decomposition, designed to enhance interclass separability and intraclass compactness by systematically capturing the hierarchical structure of geospatial relationships for fine-grained G-SGG.
2) In HPL-Net, HC with a tree-based structure is designed to enhance the distinguishability between prototypes, and CC with class-aware margins is proposed to avoid the collapse of tail-class prototype spaces caused by the imbalanced relationship categories.

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3) Our HPL-Net significantly outperforms about ten G-SGG models on the satellite-based STAR and drone-based AUG datasets across predicate classification (PredCls), scene graph classification (SGCls), and scene graph detection (SGDet) tasks. Also, the effectiveness of key components is verified by extensive ablation and visualization analysis.

II. Related Work

A. Visual Relationship Understanding

SGG has attracted significant attention since the release of large-scale datasets for SGG in natural imagery [18], [19], [20], [21] and G-SGG in RSI [22], [23], [24]. Unlike conventional detection and segmentation tasks [25], [26], [27], which treat objects as isolated entities, SGG explicitly models the relationships between objects within an image, enabling a deeper understanding of the scene. To enhance SGG performance, numerous tasks [28], [29], [30], [31], [32] have been proposed to solve the SGG task from different perspectives. Xu et al. [20], Li et al. [33] enhance the contextual modeling capabilities of models by encoding the contextual information of objects and edges. Subsequently, various debiasing SGG frameworks [34], [35], [36] have been proposed to address biases in relationship understanding. To address the ambiguity between coarse-grained and fine-grained predicate annotations in SGG datasets, several studies [37], [38] have adopted a coarse-to-fine division of relationship predicates, combined with hierarchical or grouped distillation strategies, thereby improving the accuracy of relationship classification. Moreover, commonsense including language priors [18], statistical priors [39], [40], and external knowledge [41], [42], [43], [44] can be regarded as a type of experience that enables networks to accurately understand the relationships within a scene. Based on external knowledge introduced from large text corpora, PENet [8] modeled prototypes of the entities and relationships in the semantic space, thereby generating compact and discriminative representations for relationship understanding. However, in geospatial relationship understanding, the prototype corresponding to each relationship type is not unique and varies with changes in subjects and objects. In other words, the number of prototypes for each category should adapt to the intraclass variability within the category.

B. Prototype Learning and Metric Learning

Metric learning involves designing a distance metric to cluster samples of the same category together while pushing apart samples of different categories [45], [46]. Closely related to this, prototype learning aims to learn discriminative class prototypes, encouraging samples within the same class to be close to their prototype and well-separated from other classes in the metric space [47]. Owing to its exceptional ability to learn discriminative representations, integrating metric learning into prototype-based networks has demonstrated clear advantages for various tasks in computer vision [48], [49], [50]. In RSI, prototype learning has attracted increasing attention, particularly for few-shot tasks. For image classification, Liu et al. [10] proposed a category-specific prototype self-refinement contrastive learning method that uses supervised contrastive learning to promote intraclass compactness and interclass dispersion of features in the metric space. For semantic segmentation, Zhu et al. [51] proposed an unsupervised domain adaptation network for semantic segmentation, incorporating invariant domain-level prototype memory. Sun et al. [52] proposed a semantic segmentation algorithm for RSI based on nonlearnable prototypes and clustering strategies to address foreground–background imbalance and intraclass variance. For object detection, Ni et al. [53] introduced a prototype learning network with proposal relation for detecting small targets in complex backgrounds, and Ouyang et al. [54] proposed a prototypical contrastive learning-based detector for fine-grained object detection. However, most of these approaches model each class with a single prototype, which may limit their capacity to capture the inherent diversity of visual features in RSI. To address this, recent works have explored multiprototype and hierarchical representations. Liu et al. [55] proposed multilevel prototype alignment for cross-domain few-shot classification. Gao et al. [56] introduced a multiprototype rectification strategy to address large intraclass variation, while Shen et al. [57] developed an adaptive self-supporting prototype learning network that assigns multiple prototypes to improve segmentation performance. Despite these advances, there remains limited research on whether hierarchical structures of geospatial relationship prototypes can enhance fine-grained G-SGG in RSI. To address this gap, in this work, we propose hierarchical relationship prototypes for fine-grained G-SGG, modeled through upward aggregation and downward decomposition. The effectiveness of these hierarchical prototypes is demonstrated through visualization experiments, offering new insights into their role in enhancing geospatial relationship understanding.

III. Methodology

As shown in Fig. 2, HPL-Net follows the standard pipeline commonly adopted in SGG methods [8], [24], [38], which consists of object detection and relationship prediction. The detector is a pretrained object detector that supports both horizontal and oriented bounding boxes. Given an input RSI, the detector generates a set of object representations, including visual features, initial object labels, and spatial features derived from bounding box coordinates. To facilitate fine-grained geospatial relationship understanding in G-SGG, HPL-Net introduces three key modules. Specifically, hierarchical prototypes are first constructed and initialized through an up-aggregation and down-decomposition of relationship prototypes. The relationship representations are then further refined through a prototype-based dictionary, which is continuously updated in hierarchical prototype learning. Finally, the hierarchical prototypes and relationship representations undergo continuous matching and optimization via HC and CC. HC with a tree-based structure mitigates prototype space overlap by enhancing the distinguishability between prototypes, while

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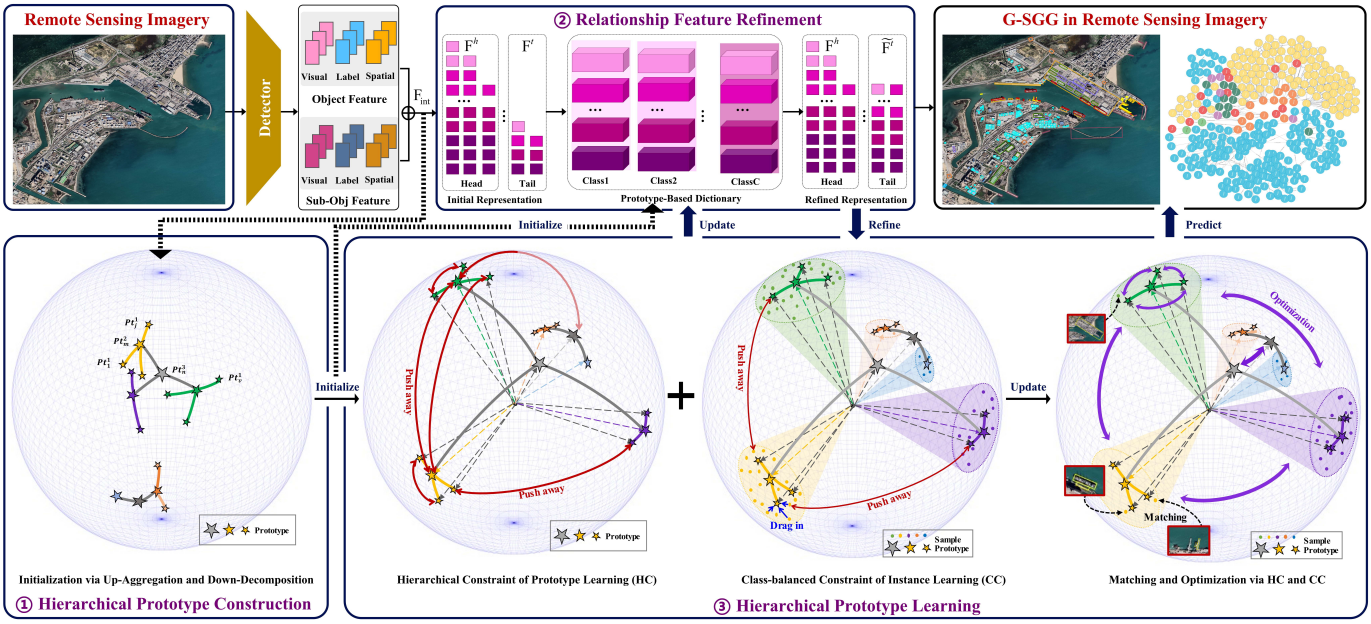


Fig. 2. Work flow of HPL-Net. First, hierarchical prototypes are initialized by up-aggregating and down-decomposing relationship prototypes. The relationship representations are then refined through a prototype-based dictionary, which is continuously updated by hierarchical prototype learning. Finally, hierarchical prototypes and relationship representations are optimized via HC with a tree structure and CC with class-aware margins.

CC with class-aware margins mitigates tail-class misclassification caused by the imbalance of relationship categories by expanding the tail-class prototype space.

A. Hierarchical Prototype Construction

In the initialization of hierarchical prototypes, HPL-Net adaptively clusters relationship prototypes upward to obtain coarse-grained parent prototypes and decomposes them downward to derive fine-grained child prototypes. Specifically, we initialize the bottom-layer prototypes by analyzing the semantic variability of subject-object combinations associated with each relationship category. For each relationship category, all subject-object combinations observed in the training set are collected. Each subject and object entity is individually encoded using pretrained word embeddings [58], and the resulting vectors are concatenated to form a joint representation for each subject-object pair.

To quantify the intraclass diversity of a given relationship prototype Pt_i, we compute pairwise cosine similarity among all subject-object vectors associated with Pt_i. Vectors with similarity exceeding a predefined threshold δ_1 are iteratively merged, yielding new provisional vectors. This process continues until all remaining vectors exhibit similarity below δ_1, indicating sufficiently distinct semantics. The final number of clusters, denoted as k_i, represents the number of fine-grained child prototypes associated with the relationship prototype Pt_i. When Pt_i corresponds to a unique subject-object combination, its child prototype count is set to $k_i = 1$ by default. By applying this procedure across all C relationship categories, we obtain the child prototype counts $\{k_1, k_2, \ldots, k_C\}$, which together determine the total number of child prototypes $K^1 = \sum_{i=1}^{C} k_i$. At the second layer, we explicitly set the prototype count $K^2 = C$ to align with the original number of relationship categories. Each of these C relationship prototypes at this layer is represented by the average vectors of all child prototypes associated with the category. To obtain higher layer parent prototypes, we compute the pairwise cosine similarities between these C vectors. Vectors with similarity exceeding a threshold δ_2 ($\delta_2 < \delta_1$) are iteratively merged, and the resulting number of distinct clusters determines the number of parent prototypes K^3. This process is recursively applied, yielding a hierarchical prototype structure with prototype counts $K = \{K^1, K^2, \ldots, K^L\}$. Based on this initialization, HPL-Net progressively updates the hierarchical prototype representations via bottom-up K-means clustering [59]. At each layer l, the prototypes Pt^l are clustered into K^{l+1} groups, and their centroids are used as parent prototypes Pt^{l+1}.

B. Relationship FR

The features for the subject, object, and subject-object pairs form the input for G-SGG models, significantly impacting the performance of relationship understanding [60]. Notably, multimodal features (such as visual, label, and spatial features) all contribute to understanding relationships from different perspectives. To predict relationships more accurately in RSI, we employ multimodal extraction [22] to obtain both object and subject-object features, which are then fused to form the initial relationship representations F_{int}.

To mitigate the prediction bias caused by the long-tail distribution of relationships, HPL-Net generates high-quality refined representations for tail-class relationships by leveraging a prototype-based dictionary $\mathbf{D} = \{Pt^l|_{l=1}^{L}\}$, where Pt^l denotes the prototypes at the lth layer and L is the total number of layers. In detail, the initial relationship representations F_{int}

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are divided into head-class F^h and tail-class F^t according to their category frequency rank. Given a random tail-class feature, HPL-Net utilizes a prototype selector, constructed using fully connected layers and a softmax function, to predict the attention map $\mathbf{M}$, which captures the contextual similarity between F^t and all prototypes stored in the dictionary. By synthesizing prototypes with learned attention, the residual representation $\mathbf{r}^t = \mathbf{MD}$ is formed. The refined tail-class representation $\widetilde{F^t}$ is then produced by $\widetilde{F^t} = F^t + \mathbf{r}^t$. Finally, the refined relationship representation Rel is obtained by adding $\widetilde{F^t}$ to the initial F_{int}.

C. Hierarchical Prototype Learning

Hierarchical prototype learning improves the interclass separability and intraclass compactness of relationship representations by coupling HC with a tree structure and CC with class-aware margins. To obtain representations of relationships and prototypes in the same space, the tree-structured prototype representations Pt and the refined relationship representations Rel are updated in the same space by $\mathcal{P} = En(Pt)$ and $\mathcal{R} = En(\text{Rel})$, where $En(\cdot)$ denotes two-layer MLPs.

1) HC of Prototype Learning: Prototype learning aims to obtain discriminative prototype representations in the latent space and avoid prototype space overlap. Given any relationship prototype, previous research often treats all other relationship prototypes as negative samples, indiscriminately pushing them equally away. However, this neglects the varying correlations between relationship categories, which may lead to over-penalization for categories with stronger correlations. For example, in the case of nonoverlapping prototype spaces, triplet <chimney, ve, and smoke> should be pushed farther away by triplet <ship, away from, and dock> than triplet <ship, approach, and dock> by triplet <ship, away from, and dock>.

To mitigate such over-penalization, we propose the HC strategy with a tree structure, with its computation intuitively illustrated in Algorithm 1. Specifically, under the HC framework, given a relationship prototype $\mathcal{P}_i^l$ at the lth layer, its negative set includes all other prototypes from the same layer, as well as nonparent prototypes from the $(l+1)$th layer. To this end, we compute the cosine similarity between each prototype and its negative prototypes in a bottom-up manner, thereby enforcing hierarchical repulsion on the hypersphere. The HC loss $\mathcal{L}_{\text{HC}}$ is defined as

$$\mathcal{L}_{\text{HC}} = \frac{1}{N}\sum_{i=1}^{K}\sum_{\mathcal{P}_j\in\mathcal{N}(\mathcal{P}_i^l)} \log\left(1+\exp\left(\frac{\mathcal{P}_i^l\cdot\mathcal{P}_j}{\tau}\right)\right) \tag{1}$$

where $N = \sum_{i=1}^{K}|\mathcal{N}(\mathcal{P}_i^l)|$ denotes the total number of negative prototypes across all prototypes, and $K = \sum_{l=1}^{L}|\mathcal{P}^l|$ represents the total number of prototypes across all layers. τ is a temperature parameter. $\mathcal{N}(\mathcal{P}_i^l)$ denotes the set of negative prototypes for $\mathcal{P}_i^l$, including both other prototypes from the lth layer and nonparent prototypes from the $(l+1)$th layer.

Algorithm 1 HC of Prototype Learning

Require: Tree-structured prototype set $\{\mathcal{P}^l\}_{l=1}^{L}$, and the temperature parameter τ.
Ensure: Hierarchical constraint loss $\mathcal{L}_{HC}$.

```
L_HC ← 0, N ← 0
for each layer l = 1 to L do
    for each prototype P_i^l ∈ P^l do
  // * Collect all negative prototypes. *//
        N^l ← {P_j^l | j ≠ i}
        N^{l+1} ← {P_u^{l+1} | u ∉ Parent(i)}
        N(P_i^l) ← N^l ∪ N^{l+1}
  // * Compute cosine similarity. *//
        for each P_j ∈ N(P_i^l) do
            s_ij ← cos(P_i^l, P_j)
            L_HC ← L_HC + log(1 + exp(s_ij/τ))
        end for
        N ← N + |N(P_i^l)|
    end for
end for
L_HC ← L_HC/N
```

2) CC of Instance Learning: The gist of instance-wise matching is to embed samples in the prototype space, bringing them close to their corresponding prototypes as positive samples and far from others as negative samples, achieving compact and distinguishable relationship representations. However, for long-tailed data distributions, head-classes constrain the spatial extent of tail-classes, preventing the model from covering the true distribution of tail-class relationships with limited training samples. The imbalance in relationship samples during training causes the prototype space of tail-classes to collapse into smaller regions, leading to misclassification during testing.

In this article, we propose the CC strategy with a class-aware margin factor m_c to learn more discriminative prototypes. Specifically, given the ith relationship representation $\mathcal{R}_c^i$ from category c, CC looks up the corresponding positive prototype $\mathcal{P}_c^l$, and then minimizes the distance d_c^+ between $\mathcal{R}_c^i$ and $\mathcal{P}_c^l$. Similarly, CC looks up the negative prototypes $\mathcal{P}_{c\prime}^l$, and maximizes the distance d_c^- between $\mathcal{R}_c^i$ and $\mathcal{P}_{c\prime}^l$. Accordingly, the CC loss $\mathcal{L}_{\text{CC}}$ is formulated as

$$\mathcal{L}_{\text{CC}} = \max\left(0, d_c^+ - d_c^- + m_c\right) \tag{2}$$

where $d_c^+ = \|\mathcal{R}_c^i - \mathcal{P}_c^l\|_2^2$ and $d_c^- = \|\mathcal{R}_c^i - \mathcal{P}_{c\prime}^l\|_2^2 (c \neq c\prime)$. m_c balances the prototype space by assigning a larger margin to the tail-class representation than the head-class

$$m_c = \Delta \times \frac{(1/n_c)^\beta}{\sum_{i=1}^{C}(1/n_i)^\beta} \tag{3}$$

where Δ and β are constants. n_c represents the number of samples belonging to the cth relationship category.

To address the challenge posed by the long-tailed distribution, we adopt focal loss [65] for both object and relationship

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classification, following the parameter settings in the work [66]. Formally, the focal loss is defined as

$$L_{\text{focal}} = -\alpha (1-p)^{\gamma} \log (p) \tag{4}$$

where p is the classification score, α denotes a balancing factor to address class imbalance, and γ is a focusing parameter that down-weights well-classified examples to emphasize hard negatives.

Finally, during the training stage, the overall loss $\mathcal{L}_{\text{G-SGG}}$ is expressed as

$$\mathcal{L}_{\text{G-SGG}} = \mathcal{L}_{\text{focal}} + \mathcal{L}_{\text{HC}} + \mathcal{L}_{\text{CC}}. \tag{5}$$

IV. Experiments

A. Experiment Datasets

1) STAR Dataset[1] [22]: SGG in satellite imagery (STAR) contains more than 210 000 objects and over 400 000 geospatial relationships across 1273 complex scenarios worldwide, with image sizes ranging from 512 × 768 to 27 860 × 31 096 pixels. This article followed the strategy [22] to split the dataset (i.e., 60%/20%/20% images for train/val/test, 48 object and 58 relationship categories).

2) AUG Dataset[2] [67]: Aerial image urban SGG (AUG) consists of 400 aerial images, each with a format size of 4000 × 6000 pixels. The dataset includes 25 594 objects and 16 970 relationships. Consistent with the setup in work [67], we utilize the AUG dataset with a split of 360 images for training and 38 images for testing.

B. Experiment Settings

1) Implementation Details: For the STAR dataset, we utilize the HOD-Net and PPG network [22] to detect objects and obtain subject-object pairs in large-scale RSI. The batch size and initial learning rate are set to 4 and 1×10^{-3} for the PredCls and SGCls tasks, and 2 and 1×10^{-3} for the SGDet task, respectively. For the AUG dataset, following prior works [67], Faster R-CNN [68] is used as the detector, with a batch size of 2 and an initial learning rate of 5×10^{-3} consistently applied across the PredCls, SGCls, and SGDet tasks. Specifically, we perform object detection for the SGDet task using a per-class NMS with 0.5 IoU [34]. The hyperparameters Δ and β are set to 100 and 0.125, respectively, and the number of hierarchical layers L is set to 3. The thresholds δ_1 and δ_2 are empirically chosen as 0.9 and 0.5, respectively. The focal loss parameters α and γ are set to 0.1 and 2, respectively, following the settings in [66]. All experiments were conducted using a single NVIDIA GeForce RTX 3090 GPU (24 GB).

2) Tasks Settings: Three standard evaluation tasks are used below.

1) *Predicate Classification:* Predict relationship labels given ground-truth (GT) subject and object boxes and labels.
2) *Scene Graph Classification:* Predict subject, object, and relationship labels given GT subject and object boxes.
3) *Scene Graph Detection:* Predict all three labels and two boxes given RSI.

[1]https://linlin-dev.github.io/project/STAR.html
[2]https://drugd.github.io/AUGDataset.github.io/

3) Evaluation Metrics: To ensure a fair evaluation across all relationship categories, we report the performance on mean multilabel recall@K (mMR@K) for the STAR dataset. Moreover, we utilize head multilabel recall@K (HMR@K) and tail multilabel recall@K (TMR@K) to assess the effectiveness of G-SGG methods on head and tail relationships, with K set to 1500/2000. For the AUG dataset, following the previous works [67], [69], R@K, mR@K, and F@K are chosen as evaluation metrics, where F@K is the harmonic average of R@K and mR@K, and K is set to 50/150.

4) Model Zoo: We evaluated approximately ten representative G-SGG models, including two specifically designed for RSI: RPCM [22] and LPG [67]. In addition, we included widely adopted models originally developed for natural imagery, such as IMP [20], Motif [39], GPS-Net [61], SHA [38], HETSGG [63], PE-Net [8], as well as improvement models, such as RTPB (Motif) [35], CFA (Motif) [62], and SBP (Motif) [64], among others. These models have been integrated into a unified framework as established in prior works [22], [67], making them applicable to G-SGG tasks and forming a standardized benchmark for evaluation. In our experiments, following the same protocol, all models were evaluated on the same dataset using a unified codebase and a shared pretrained detector backbone to ensure a fair comparison.

C. Comparison With the State-of-the-Art Methods

1) Experiments on the STAR Dataset: Table I gives a comparison of our HPL-Net with other state-of-the-art methods for the mMR@K, HMR@K, and TMR@K metrics on the PredCls, SGCls, and SGDet tasks. Due to the SGDet task being more susceptible to the influence of object misclassification by detectors, our HPL-Net shows a more significant advantage over other methods in PredCls and SGCls tasks. In detail, our HPL-Net outperforms the prototype-based PE-Net by 5.54%/5.76%, 10.01%/9.76%, 3.14%/3.56%, and recent RPCM by 1.29%/1.75%, 0.61%/0.51%, and 0.36%/0.79% at mMR@1500/2000 on three subtasks. In addition, while HPL-Net exhibits a slight decrease in HMR@K on the PredCls and SGCls tasks, this is compensated by a more significant improvement in TMR@K, which better captures the model's effectiveness in tail-class relationship prediction. For the TMR@1500/2000 metric, HPL-Net outperforms PE-Net by 10.65%/10.66%, 15.99%/15.39%, 1.53%/1.17%, and RPCM by 4.9%/5.32%, 1.79%/1.56%, 0.4%/0.12% on all subtasks, respectively. Table II presents the inference time comparison between HPL-Net and representative state-of-the-art G-SGG methods for the PredCls task on the same RTX 3090 GPU. While achieving the highest mMR@2000 of 44.05%, HPL-Net demonstrates a clear efficiency advantage, requiring only 1.44 s per image, which is 17.2% faster than RPCM and close to the fastest baseline. The total inference time of 380.50 s for the entire test set further confirms its capability to process large-scale datasets within practical time constraints. These results demonstrate the effectiveness of HPL-Net in achieving fine-grained geospatial relationship understanding within satellite-based G-SGG tasks.

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TABLE I

QUANTITATIVE COMPARISONS (%) WITH STATE-OF-THE-ART METHODS ON THE SATELLITE-BASED STAR TEST SET FOR THE PREDCLS, SGCLS, AND SGDET TASKS. BOLD INDICATES THE BEST RESULTS, AND UNDERLINING DENOTES THE SECOND-BEST RESULTS

Models	PredCls			SGCls			SGDet		
	mMR @1500/2000	HMR @1500/2000	TMR @1500/2000	mMR @1500/2000	HMR @1500/2000	TMR @1500/2000	mMR @1500/2000	HMR @1500/2000	TMR @1500/2000
IMP [20]	21.47/22.31	28.86/30.19	14.08/14.43	18.57/19.38	24.79/25.82	12.35/12.94	5.60/5.99	9.13/9.51	2.07/2.47
Motif [39]	29.40/30.64	36.73/38.02	22.07/23.26	24.21/25.36	28.57/30.45	19.85/20.27	7.78/8.22	11.63/12.23	3.93/4.21
GPS-Net [62]	34.08/35.32	**41.63/43.23**	26.53/27.41	14.28/14.90	16.89/17.84	11.67/11.96	9.54/10.09	14.01/14.51	5.07/5.67
RTPB (Motif) [35]	37.54/39.15	39.12/41.09	35.68/37.21	29.56/30.63	31.01/32.47	28.10/28.79	7.97/8.49	11.96/12.65	3.82/4.33
SHA [38]	33.26/34.78	38.06/39.98	28.46/29.58	23.30/24.15	28.51/29.48	18.09/18.82	10.18/10.79	15.85/16.46	4.51/5.12
CFA (Motif) [63]	29.98/32.83	37.16/38.73	22.80/26.93	27.33/28.39	30.26/31.81	24.40/24.97	7.85/8.29	10.94/11.47	4.76/5.11
HETSGG [64]	29.74/30.98	38.01/40.03	21.47/21.93	12.18/12.78	15.01/15.67	9.35/9.89	5.44/5.73	8.95/9.44	1.93/2.02
PE-Net [8]	36.99/38.29	38.83/40.36	35.15/36.22	20.64/21.60	26.65/28.22	14.63/14.98	8.75/9.30	11.39/11.81	6.11/6.79
SBP (Motif) [65]	33.56/34.78	38.53/40.53	28.59/29.03	28.72/29.64	31.41/32.16	26.30/27.12	7.99/8.50	9.83/10.06	6.15/6.94
RPCM [22]	41.24/42.30	41.58/43.04	40.90/41.56	30.04/30.85	**31.65/32.89**	28.43/28.81	11.53/12.07	15.82/16.27	7.24/7.87
HPL-Net (Ours)	**42.53/44.05**	39.27/41.22	**45.80/46.88**	**30.65/31.36**	31.08/32.35	**30.22/30.37**	**11.89/12.86**	**16.06/17.73**	**7.64/7.99**

TABLE II

ACCURACY AND INFERENCE TIME COMPARISON BETWEEN HPL-NET AND OTHER G-SGG METHODS FOR THE PREDCLS TASK ON THE SAME RTX 3090 GPU, WHERE TIME REPRESENTS THE PER-IMAGE INFERENCE TIME (SECONDS) PER DEVICE, AND TOTAL TIME REPRESENTS THE TOTAL INFERENCE TIME (SECONDS) FOR ALL IMAGES PER DEVICE

Model	mMR@2000 (%)	Time/img (s)	Total Time (s)
HETSGG [64]	30.98	1.47	387.59
PE-Net [8]	38.29	1.11	292.88
SBP (Motif) [65]	34.78	1.00	265.16
RPCM [22]	42.30	1.74	460.12
HPL-Net (Ours)	44.05	1.44	380.50

Fig. 3 presents qualitative visualizations of various models on two challenging RSI examples from the STAR dataset. In Fig. 3(a), the primary challenge lies in distinguishing between <cooling_tower, ve, and vapor> and <cooling_tower, se, and vapor>, which share similar visual appearances but correspond to distinct relationship categories. This requires the model to possess strong interclass separability, even under fine-grained appearance variations. The results show that models, such as IMP, HETSGG, and PE-Net, fail to classify these relationships accurately, while RPCM and our proposed HPL-Net correctly differentiate between ve and se with higher precision. This performance gain is primarily attributed to the strong interclass separability enabled by prototype-based learning, particularly in HPL-Net, which further enhances the discrimination of fine-grained relationships. Another challenging example is shown in Fig. 3(b), which involves distinguishing parking status between vehicles and parking lots, as well as identifying the movement trend between vehicles and toll_gates. This scenario demands a model to capture finer details, such as a vehicle's head. While models like IMP, PE-Net, HETSGG, and RPCM perform well on clearly distinguishable relationships, they exhibit varying degrees of misclassification when confronted with such minor visual differences. In contrast, our HPL-Net not only successfully identifies incorrect parking status but also accurately distinguishes between <car, away from, and toll_gate> and <car, approach, and toll_gate>. This demonstrates that HPL-Net, by leveraging hierarchical structural constraints, can effectively capture fine-grained semantic variations among relational entities. Specifically, the prototype-based hierarchical modeling enables HPL-Net to capture intraclass variations and preserve interclass boundaries, leading to more precise relationship understanding. Overall, HPL-Net demonstrates a clear advantage in learning discriminative relationship representations. Even when predictions are not annotated in the GT, our results remain semantically meaningful, as exemplified by the GT <car 0, in the same parking with/ parked alongside with, car 1> versus predicted <car 0, within safe distance of, car 1>, and the GT <car 4, in the different parking with, car 6> versus the predicted <car 4, within safe distance of, car 6>.

2) Experiments on the AUG Dataset: To evaluate the generalizability of our method across different datasets, we conduct experiments on the drone-based AUG dataset, with the results presented in Table III. Generally, our method

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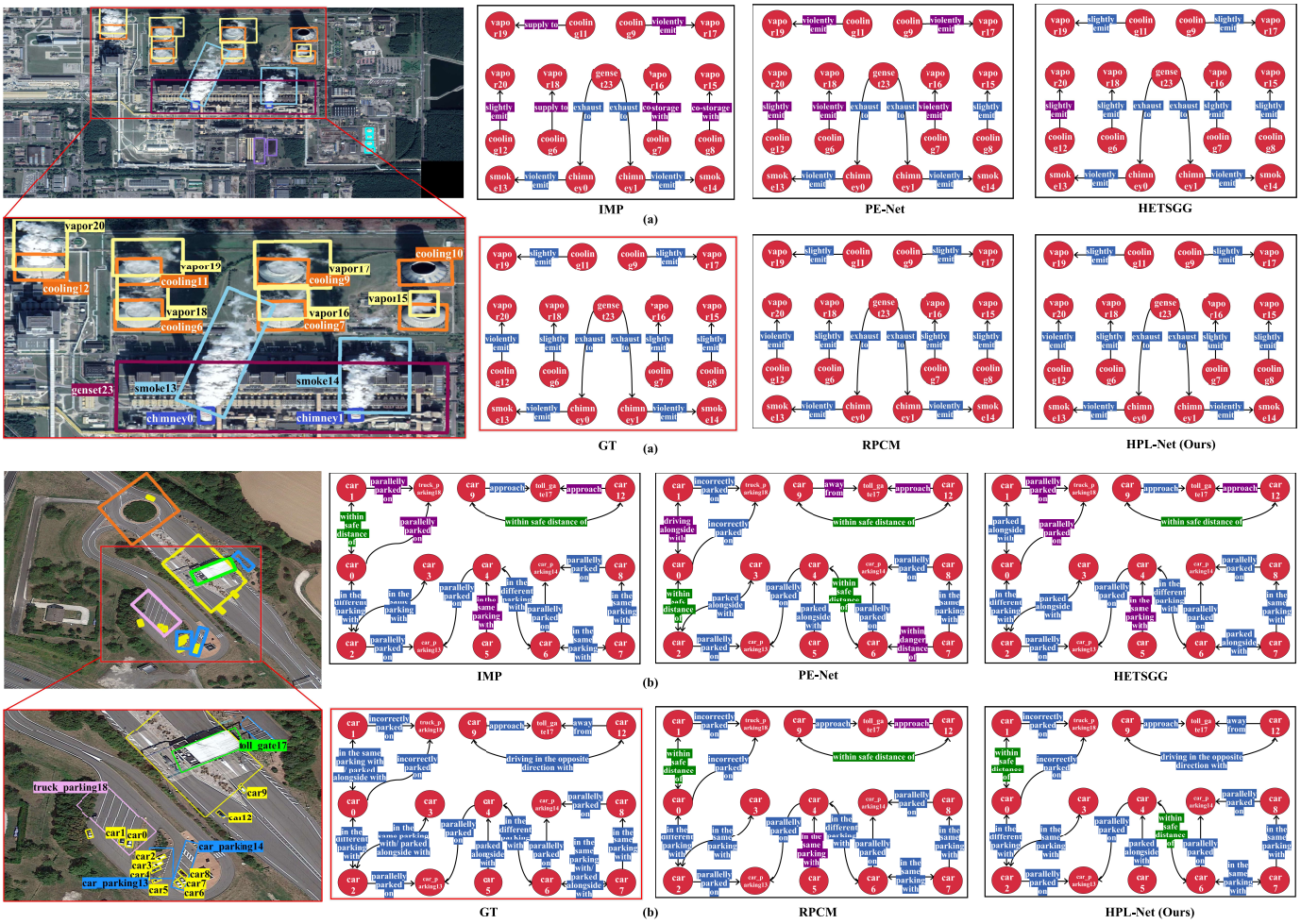


Fig. 3. Visualization results of G-SGG on the satellite-based STAR dataset. Blue edges mean the correctly predicted relationships, purple edges are GT relationships that failed to be detected, and green edges indicate reasonable relationships predicted by the model but not annotated in the GT of (a) and (b).

achieves superior performance compared to other SGG models. Due to the SGDet task being more susceptible to the object detector, our method demonstrates a more significant advantage over other methods for the PredCls and SGCls tasks. Concretely, our HPL-Net outperforms the latest LPG by 5.92%/4.31%, 3.12%/1.10, and 0.11%/0.10% on mR@50/150 and by 4.52%/4.31%, 4.70%/1.82, and 0.53%/0.21% on F@50/150 for the PredCls, SGCls, and SGDet tasks. In addition, it also outperforms the prototype-based SGG model, PENet, by 7.23%/6.41%, 8.87%/7.28%, and 1.70%/2.07% at F@50/100 on PredCls, SGCls, and SGDet, respectively. This robustly confirms HPL-Net's ability to generalize across diverse data distributions in fine-grained geospatial relationship understanding tasks.

Fig. 4 demonstrates the generalization performance of HPL-Net on the UAV-based AUG dataset. As shown in Fig. 4(a), a challenging case involves recognizing the interaction between two individuals from an aerial perspective, which necessitates the capture of fine-grained pose features by the model. HPL-Net successfully predicts the relationship <woman1, play with, and woman2>, owing to its ability to model subtle intraclass variations through hierarchical prototype learning. Another difficult example is illustrated in Fig. 4(b), where the model must identify detailed interactions between a person and a bicycle, such as hand and leg positions in contact with the bike. In this case, HPL-Net accurately detects the relationship <man49, ride, and bicycle2>, benefiting from its ability to model fine-grained visual cues through hierarchical prototype representation. Overall, the cross-dataset generalization experiment demonstrates that HPL-Net is capable of producing results consistent with GT annotations. Even in cases where GT annotations are absent, it can generate semantically meaningful predictions, such as <man49, near, and car47> and <car47, be parked beside, and grassland4>.

D. Ablation Studies

1) Effect of Hierarchical Layers: To further explore the impact of hierarchical layer L on geospatial relationship understanding, Table IV presents the performance of our HPL-Net on the PredCls task with the number of layers set to 2, 3, 4, and 5. The results demonstrate that as L increases, the overall accuracy initially improves and then declines, with the best performance achieved at $L = 3$.

2) Effect of Different Components: To assess the contribution of each component in the proposed HPL-Net, we perform

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TABLE III

QUANTITATIVE COMPARISONS (%) WITH STATE-OF-THE-ART METHODS ON THE DRONE-BASED AUG TEST SET FOR THE PREDCLS, SGCLS, AND SGDET TASKS. BOLD INDICATES THE BEST RESULTS, AND UNDERLINING DENOTES THE SECOND-BEST RESULTS

Models	PredCls			SGCls			SGDet		
	R@ 50/150	mR@ 50/150	F@ 50/150	R@ 50/150	mR@ 50/150	F@ 50/150	R@ 50/150	mR@ 50/150	F@ 50/150
Motif [39]	26.40/35.80	11.30/15.80	15.83/21.92	20.20/25.00	5.10/6.90	8.14/10.82	12.80/16.20	3.80/4.70	5.86/7.29
VCTree [71]	29.10/37.70	12.50/15.80	17.49/22.27	22.00/26.60	5.20/7.60	8.41/11.82	11.50/13.40	3.10/3.70	4.88/5.80
GPSNet [62]	28.20/37.20	12.00/16.50	16.84/22.86	17.20/22.50	6.50/9.10	9.43/12.96	13.10/17.10	4.60/5.40	6.81/8.21
TDE (Motif) [34]	33.00/43.20	11.00/14.60	16.50/21.82	20.60/25.20	5.20/6.30	8.30/10.08	12.70/15.50	3.00/4.50	4.85/6.98
TDE (VCTree) [34]	33.70/44.70	11.30/18.10	16.92/25.77	22.80/26.70	4.80/6.20	7.93/10.06	13.90/17.60	3.30/4.40	5.33/7.04
EBM (Motif) [72]	31.00/41.70	10.50/14.60	15.69/21.63	18.30/23.10	3.20/5.10	5.47/8.36	11.6/14.30	2.10/2.70	3.56/4.54
DLFE (Motif) [73]	25.90/36.20	9.10/15.10	8.52/21.31	15.90/21.10	3.20/4.80	5.33/7.82	11.8/14.1	4.00/4.40	5.97/6.71
IETrans (Motif) [74]	22.90/30.70	11.40/14.50	15.22/19.70	2.80/3.40	1.10/1.40	1.58/1.98	0.70/1.40	0.20/0.30	0.31/0.49
PENet [8]	30.95/40.05	13.23/18.93	18.54/25.71	16.63/22.25	3.87/5.69	6.28/9.06	12.52/16.72	3.73/5.42	5.75/8.19
LPG [68]	35.30/49.10	14.20/19.40	21.25/27.81	19.80/28.00	7.10/9.80	10.45/14.52	14.00/22.10	4.60/6.50	6.92/10.05
HPL-Net (Ours)	**35.91/49.80**	**20.12/23.71**	**25.77/32.12**	**29.41/32.60**	**10.22/10.90**	**15.15/16.34**	**17.9/23.0**	**4.71/6.60**	**7.45/10.26**

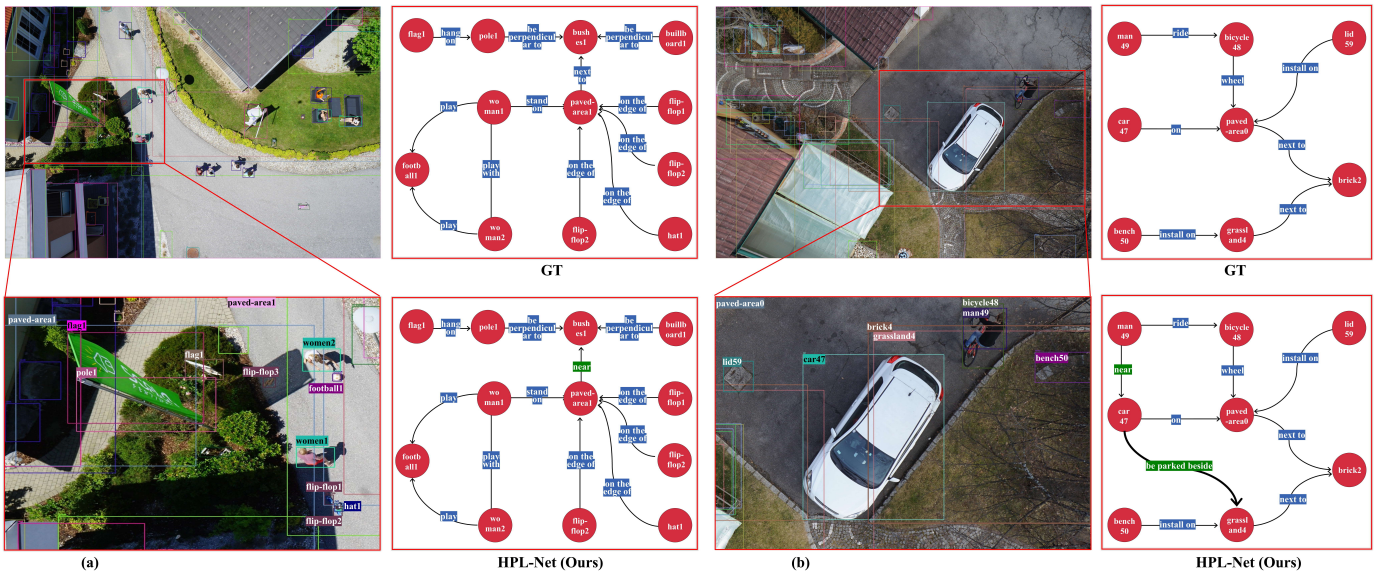


Fig. 4. Visualization results of G-SGG on the drone-based AUG dataset. Blue edges mean the correctly predicted relationships, and green edges indicate reasonable relationships predicted by the model but not annotated in the GT of (a) and (b).

ablation studies on hierarchical prototype-based feature refinement (FR), HC, and CC. To minimize the influence of object misclassification by the detectors, we focus exclusively on the PredCls task for our experiments. The corresponding results are presented in Table V. Not surprisingly, the experimental results show that FR, HC, and CC modules are important for the performance of G-SGG. Concretely, Line 1, HPL-Net is trained without FR, which directly uses HC and CC to classify the relation feature. Line 2 and Line 3, HPL-Net is trained with FR, which only uses HC or CC to predict relationship types. Line 4, HPL-Net is trained with FR, HC, and CC, enabling it to effectively capture the hierarchical structure of relationships and generate more discriminative representations. Fig. 5 illustrates the impact of the FR module on different relationship categories. It is evident that incorporating the FR module significantly enhances the accuracy of tail categories (e.g., se," "*pass through*," and "*pass under*"), demonstrating its effectiveness in refining and improving tail-class relationships.

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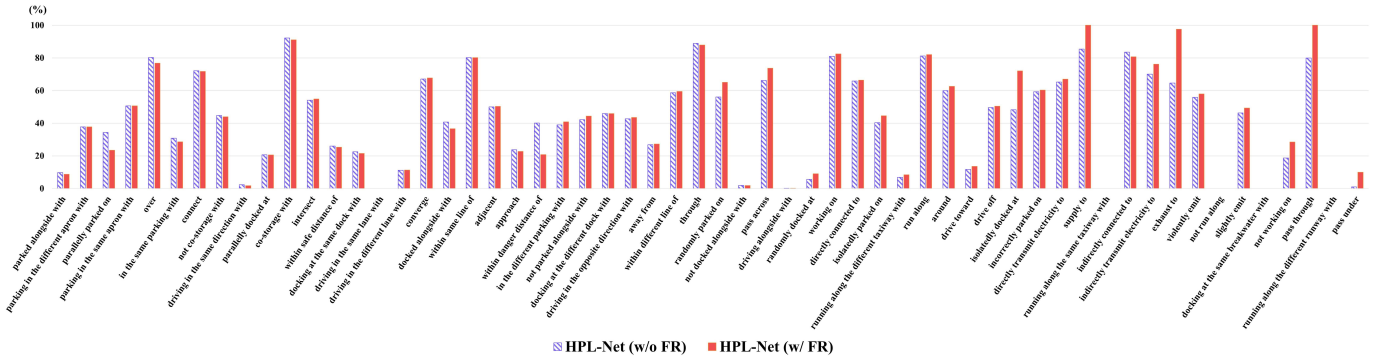


Fig. 5. R@2000 for all relationship predicates in the STAR dataset under the PredCls task between HPL-Net w/o FR and HPL-Net w/ FR. Relationship predicates are sorted by their frequencies in STAR.

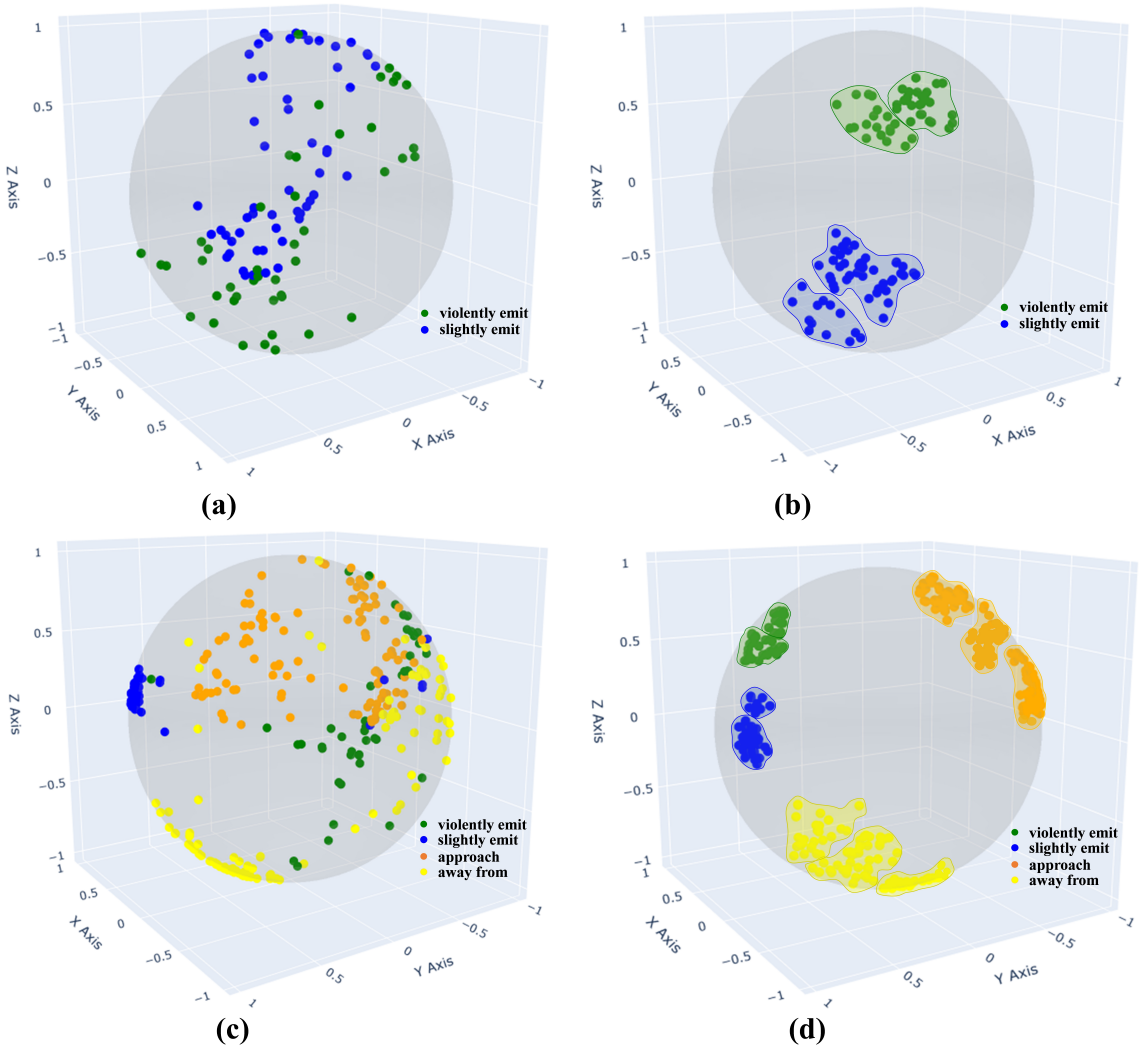


Fig. 6. Comparison of t-SNE visualizations of relationship representations from the STAR dataset on the hypersphere. (a) and (c) Distributions before applying HPL-Net, while (b) and (d) distributions after applying HPL-Net.

3) Visualization in Latent Spaces: To verify the existence of a hierarchical structure in geospatial relationships and evaluate whether HPL-Net's hierarchical structure improves fine-grained relationship understanding, we use the t-SNE technique to visualize fine-grained relationship representations (such as se, "ve," "*away from*," and "*approach*") on the hypersphere, shown in Fig. 6. As anticipated, our HPL-Net achieves high intraclass compactness and interclass separability of relationship representations. In detail, Fig. 6(a) and (c) are the initial representation of subject-object pairs before

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TABLE IV

IMPACT OF HIERARCHICAL LAYERS ON HPL-NET FOR THE PREDCLS TASK ON THE SATELLITE-BASED STAR DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINING DENOTES THE SECOND-BEST RESULTS

L	mMR @1500/2000	HMR @1500/2000	TMR @1500/2000
2	41.18/42.73	38.65/41.03	43.71/44.43
3	**42.53/44.05**	**39.27/41.22**	**45.80/46.88**
4	41.46/42.87	39.16/41.23	43.76/44.51
5	39.83/40.99	38.35/40.19	41.31/41.79

TABLE V

COMPARISON OF DIFFERENT COMPONENTS ON HPL-NET FOR THE PREDCLS TASK ON THE SATELLITE-BASED STAR DATASET. BOLD INDICATES THE BEST RESULTS, AND UNDERLINING DENOTES THE SECOND-BEST RESULTS

FR	HC	CC	mMR @1500/2000	HMR @1500/2000	TMR @1500/2000
	✓	✓	41.15/42.09	37.63/38.73	44.67/45.45
✓	✓		37.57/38.81	40.65/**42.62**	34.49/35.00
✓		✓	37.04/38.10	**40.76**/42.49	33.32/33.71
✓	✓	✓	**42.53/44.05**	39.27/41.22	**45.80/46.88**

our HPL-Net, and Fig. 6(b) and (d) are the relationship representation after our HPL-Net. As can be observed, after HPL-Net, the representations that belong to the different categories are well-separated, and those belonging to the same category are more compact. Notably, in Fig. 6(b) and (d), the samples corresponding to the relationships form multiple clusters on the hypersphere, each with varying quantities, and the distances between clusters of the same category are smaller than those between clusters of different categories. This observation further validates the effectiveness of modeling fine-grained geospatial relationships through hierarchical relationship prototypes via aggregation-decomposition. This improvement stems from two key aspects: first, HPL-Net helps the model learn discriminative relationship representations through HC, thereby differentially pushing different categories of relationship representations in the latent space. Second, HPL-Net expands the tail-class prototype space using hierarchical prototype-based FR and CC, which mitigates misclassification caused by class imbalance.

V. CONCLUSION

HPL-Net offers a robust solution for fine-grained G-SGG, demonstrating strong performance on both satellite-based and drone-based imagery. However, it operates under a closed-set vocabulary assumption, where all object and relationship categories are assumed to be known during training. A promising direction for future work is to integrate open-vocabulary or zero-shot learning techniques into the hierarchical prototype framework, enhancing the model's applicability in real-world geospatial scenarios.

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Linlin Wang received the B.S. degree in geography and information science from Henan University, Kaifeng, China, in 2019, and the M.E. degree in cartography and geographical information engineering from China University of Geosciences, Wuhan, China, in 2022. She is currently pursuing the Ph.D. degree with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan.

She has published several articles in IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. Her research interests include remote sensing scene graph generation.



Tingzhu Wang received the B.E. degree from the School of Mechanical Engineering and Electronic Information, China University of Geosciences, Wuhan, China, in 2023. He is currently pursuing the M.S. degree with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan.

He has published articles in IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and ECCV. His research interests include image scene graph generation and video scene graph generation.



Junwei Luo received the B.S. degree in remote sensing science and technology from Wuhan University, Wuhan, China, in 2023, where he is currently pursuing the M.S. degree with the School of Remote Sensing and Information Engineering.

He has published several articles in IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and CVPR. His research interests include remote sensing object detection and the remote sensing vision-language foundation model.



Yansheng Li (Senior Member, IEEE) received the B.S. degree in information and computing science from Shandong University, Weihai, China, in 2010, and the Ph.D. degree in pattern recognition and intelligent system from the Huazhong University of Science and Technology, Wuhan, China, in 2015.

From 2017 to 2018, he was a Visiting Assistant Professor with the Department of Computer Science, Johns Hopkins University, Baltimore, MD, USA. He is currently a Full Professor with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan. He has authored more than 100 peer-reviewed journal articles and conference papers, including *Nature Machine Intelligence*, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, RSE, and IJCV. His research interests include knowledge graph, deep learning, and their applications in remote sensing and big data mining.

Dr. Li was awarded the Young Surveying and Mapping Science and Technology Innovation Talent Award of Chinese Society for Geodesy, Photogrammetry, and Cartography in 2022. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and an Editorial Member of *The Photogrammetric Record*.

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